

REVIEW

Neurovascular Signaling in the Perivascular Space: Biophysical Principles and Disruption in Hypertension

Thomas A. Longden¹, Abigail Vigderman¹, Suyeon Ju¹, Dominic P. Isaacs

ABSTRACT: The brain's vasculature—from pial arteries through penetrating arterioles and the capillary bed to venous drainage—is enveloped by a narrow, compositionally elaborate perivascular space that provides the primary signaling interface between neural tissue and blood vessels. Here, we advance the principle that the geometry and composition of this space are not constant but change systematically along the vascular tree, and that perivascular space composition at these locations imposes biophysical constraints that dictate the fidelity and gain of neurovascular signaling. We argue that its narrow micro-to-nanodomain geometry, basement membrane sieving effects, and the anionic environment created by heparan sulfate proteoglycans favor small aqueous mediators—potassium, nitric oxide (NO), and adenosine—for rapid, spatially targeted blood flow control, while constraining lipophilic mediators, such as prostaglandins and epoxyeicosatrienoic acids, toward predominantly autocrine action. Reassessing glymphatic fluid flux as a proposed source of vascular mechanical input, we find that the relevant shear forces fall several orders of magnitude below the threshold required to engage canonical smooth muscle mechanotransduction, arguing against a direct mechanical role for perivascular flow. We then synthesize how hypertension degrades each layer of this system: perivascular space expansion dilutes mediator concentrations, barrier breakdown admits plasma proteins capable of sequestering signaling molecules, and inflammation dysregulates the vascular ion channels that serve as signaling effectors. Framing the perivascular space as a biophysically constrained signaling compartment helps reconcile disparate observations of mediator selectivity and fluid mechanics, identifies how its progressive disruption may contribute to cerebrovascular dysfunction associated with chronic hypertension, and may help identify novel targets for preventing or reversing cognitive decline and neurodegeneration.

Key Words: blood vessel ■ hyperemia ■ hypertension ■ neurovascular coupling ■ nitric oxide

Woven throughout the brain are hundreds of miles of blood vessels forming a continuous, anastomosing network responsible not only for delivering energy substrates to nourish the neural tissue but also for waste clearance, regulation of pH, and temperature control. Execution of these tasks is rooted in the optimized coordination of blood flow, which is continually adjusted on a moment-to-moment basis and prioritized to areas most in need. This requires vessels to be constantly apprized of the activity and needs of their local neurons, making effective neurovascular communication paramount for brain function.

This is no easy feat. With hundreds of millions of synapses and astrocytic processes per cubic millimeter,¹

extracellular space tends to be vanishingly sparse and extremely tortuous, imposing significant challenges for effective communication between cells over anything more than the shortest distances. Mechanisms have thus had to evolve to address these challenges.

This article explores our current understanding of the composition of the perivascular space (PVS) throughout the brain vasculature—ranging from pial arteries on the brain surface to veins draining the tissue, with a particular focus on how these spaces shape neurovascular signaling to achieve effective control of blood flow. Not only are mechanical forces shaped by the geometry of this space, but so also are the concentration and clearance of key signaling molecules, making the PVS a critical determinant

Correspondence to: Thomas A. Longden, Department of Pharmacology & Physiology, University of Maryland School of Medicine, 655 W Redwood St, 505 Howard Hall, Baltimore, MD 21201. Email thomas.longden@som.umaryland.edu

*A. Vigderman and S. Ju contributed equally.

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Nonstandard Abbreviations and Acronyms

AC	adenylyl cyclase
Ang II	angiotensin II
AT1R	Ang II type 1 receptor
AV	ascending venule
BBB	blood-brain barrier
BM	basement membrane
cPLA2	cytosolic phospholipase A2
CSF	cerebrospinal fluid
EC	endothelial cell
ECM	extracellular matrix
EET	epoxyeicosatrienoic acid
EP	E-prostanoid
GPCR	G protein-coupled receptor
nNOS	neuronal NO synthase
NOX2	nicotinamide adenine dinucleotide phosphate oxidase 2
NVC	neurovascular coupling
PA	Penetrating arteriole
PGE2	prostaglandin E2
PKA	protein kinase A
PKG	protein kinase G
PVM	perivascular macrophage
PVS	perivascular space
SAS	subarachnoid space
sGC	soluble guanylate cyclase
SMC	smooth muscle cell
SPARC	secreted protein acidic and rich in cysteine
TRPV4	transient receptor potential vanilloid 4
VDCC	voltage-dependent Ca2+ channels
VEGF	vascular endothelial growth factor

of the fidelity and gain of neurovascular communication. Importantly, hypertension alters every parameter of the PVS—from alterations to the basement membrane (BM) and extracellular matrix (ECM), to stiffening of vessels affecting fluid mechanics, and disruption of signaling mediator release and function of effector ion channels. Therefore, advancing understanding of the PVS as a signaling interface may lead to novel treatment options to halt or reverse the impacts of hypertension and other contributory factors to neurodegenerative conditions, ultimately enabling protection of neuronal health and function.

THE PAN-VASCULAR ARCHITECTURE OF THE PVS

Pial arteries form a ramifying and anastomosing network directing blood over the surface of the brain. Penetrating arterioles (PAs) split from pial arteries and plunge

radially into the brain,² branching periodically as they dive, giving rise to first-order capillaries that bifurcate repeatedly to create the capillary bed.³ Branches from the first-order capillary are designated second-order capillaries, and subsequent branches are labeled third-order, and so forth.⁴ After ramifying extensively to create the deep capillary bed, vessels converge again to direct blood to the venular draining network with vertically oriented ascending venules (AVs) draining blood to surface pial veins.⁵ This vessel network is a syncytium of diverse cell types (endothelial cells [ECs], smooth muscle cells [SMCs], and pericytes) and its structure has been extensively reviewed elsewhere.⁶

Generally, use of both the terms paravascular and PVS refers to anatomically defined pathways that permit movement of cerebrospinal fluid (CSF) as it travels along, into, and out of the brain beside blood vessels.^{7–9} Here, we use the term PVS to refer to the area immediately adjacent to the outermost vascular cells in its various forms as vessels transition from arteries through arterioles and capillaries to veins (Figure 1). A theme that emerges from this progression, and 1 we develop as an organizing principle, is that the geometry and composition of the PVS are not constant but change systematically along the vascular tree, and these changes will dictate biophysical constraints that favor different classes of signaling mediators at different positions. The wide, compliant, CSF-continuous space around pial arteries can accommodate diffusible and carrier-borne mediators over appreciable distances; the narrowing, increasingly fenestrated PA-adjacent PVS progressively restricts this; and the fused capillary BM, lacking a fluid-filled space altogether, effectively confines signaling to small molecules that rely on electrical transmission for relay to upstream contractile machinery.

At the Brain Surface

Above the brain lies the dura, which covers the arachnoid layer, below which is the subarachnoid space (SAS). The SAS is filled with CSF and woven throughout this space is a web-like network of trabeculae composed of collagen bundles coated by leptomeningeal cells¹⁰ that form connections with the sheath of epiplial cells¹¹ surrounding pial arteries (see below).

Beneath the SAS is the pia mater, which separates the SAS from the cortex,¹⁰ though fenestrations in the pia mater do allow for CSF exchange. Compositionally, the pia is a thin layer of fibroblast-like pial cells (also called leptomeningeal lining cells or leptomeningeal cells⁹) joined together by desmosomes and able to communicate with 1 another via gap junctions.¹⁰ At surface areas devoid of vasculature, a layer of collagen bundles sits below the pia and covers a layer of BM secreted by astrocytes and their endfeet, forming the glia limitans.¹²

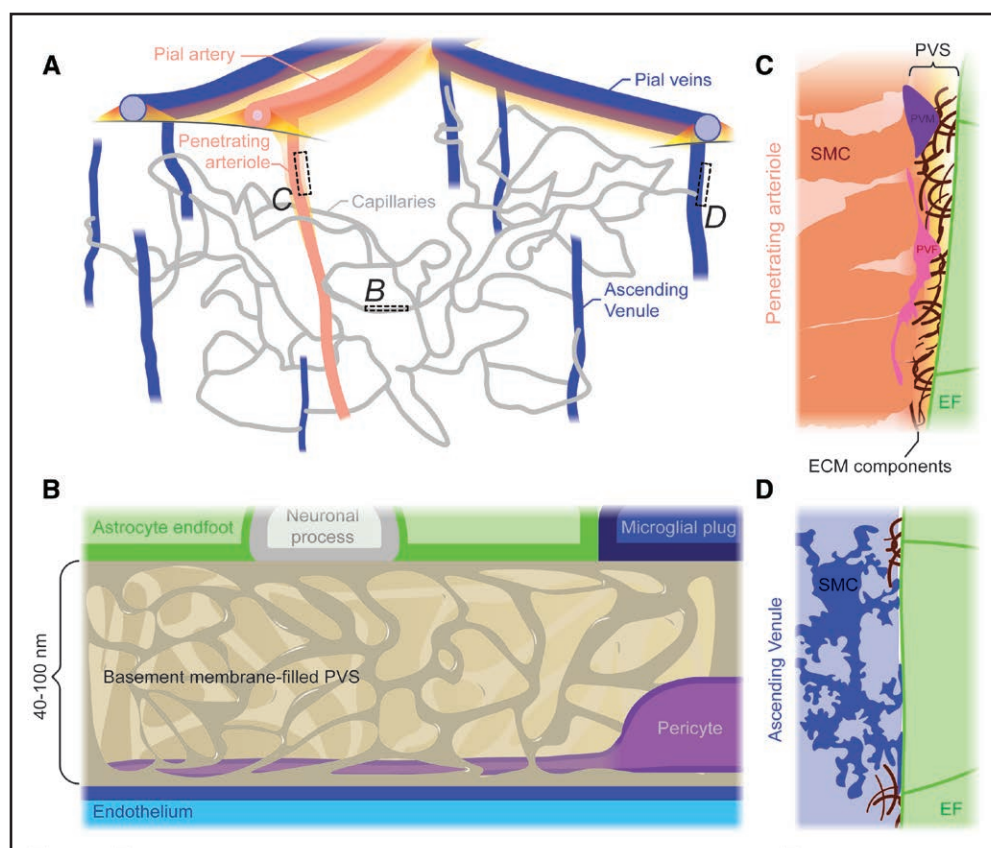


Figure 1. The shape and composition of the perivascular space (PVS) throughout the brain vasculature.

A, Brain vascular network illustrating major vessel subtypes. Fluid in the PVS is illustrated as a yellow-red translucent band adjacent to vessels, and the dimensions of this space acutely narrow with increasing depth after penetrating arterioles (PAs) branch from pial arteries and penetrate the cortex. **B**, Conceptual view of the capillary PVS corresponding to the labeled dashed box in **A**. This area is filled with basement membrane (BM) components and is <100 nm wide. Pericytes are embedded in the basement membrane, and astrocyte endfeet, neuronal processes, and microglial plugs all interface with the PVS here. **C**, At the level of PAs, the PVS is wider (tapering with depth) and contains extracellular matrix (ECM) components, perivascular fibroblasts (PVFs), and perivascular macrophages (PVMs). **D**, The PVS around ascending venules is narrower and often absent, with sparse ECM components. EF indicates endfoot; and SMC, smooth muscle cell.

The Pial Arterial and Pial Venous PVS

Pial arteries sit atop the pia mater,¹³ which reaches around to enwrap them in a circumferential sheath of reticulated epipial cells and adventitial collagen bundles. These surface vessels are somewhat isolated from the bulk CSF in the SAS¹⁴ by this pial sheath, which forms a fluid-filled paravascular channel (Table 1), but pores or “stomata” along the pial sheath lining allow for some CSF exchange.²⁰ The PVS around pial arteries thus contains CSF that has crossed the pia from the SAS, and that CSF will continue antegrade into the brain. Pial veins, in contrast, do not have a complete sheath and lack the same leptomeningeal contacts as arteries,¹³ and thus lie more freely in the CSF.

The PVS Around PAs

Around PAs, a layer of pial cells extends from the surface pia mater and covers the vessel,¹³ although its extent and occurrence across species are debated²¹ and it is thought with depth to become increasingly fenestrated

and incomplete (Figure 1; Table 1) while the width of the PVS tapers as vessels dive into the tissue. PAs are surrounded by a layer of BM which in turn is surrounded by a comprehensive astrocytic endfoot sheath that, until recently, was thought to be essentially complete.²² However, mounting evidence is challenging this principle, suggesting instead that there are periodic gaps in the endfoot coverage that allow neuronal and microglial processes into proximity with the vasculature. Indeed, capillary-associated microglia have recently been described,²³ and while not direct residents of the PVS, around 20% of these cells form small plugs with surface area $<1 \mu\text{m}^2$ between adjacent astrocyte endfeet.²⁴

Neurons also make direct contacts with intracerebral vessels. It was recently shown that glutamatergic neurons make synapse-like connections with PA SMCs, which can dilate arterioles and increase blood flow through a mechanism dependent on N-methyl-D-aspartate receptors in SMCs.²⁵ This sits alongside evidence that interneurons with processes very close to arterioles can bidirectionally modulate vessel

Table 1. PVS Dimensions and Composition Throughout the Brain Vasculature

Vessel type	Approximate PVS size		Primary composition	References
	Width	Area, μm^2		
Pial artery (brain surface)	20–60 μm	250–4500	Open fluid-filled channel.	15–18
Penetrating arteriole	n.d.	1–400	Fluid, collagen, fibroblasts, macrophages; tapers with increasing depth.	11,18
Capillary	40–100 nm*	1–2†	Fused with the basal lamina, filled with matrix components.	19
Ascending venule	n.d.	10–130	Narrow space with irregular extracellular matrix. Often absent.	17,18
Pial vein (brain surface)	n.d.	200–6000	Open fluid-filled channel.	16,18

n.d. indicates no data; and PVS indicates perivascular space.
*Values were measured manually from layer 2/3 mouse visual cortex from the MICrONS consortium data set.¹⁹
†Values are estimates calculated for an annulus using measured widths.

diameter—cells positive for vasoactive intestinal peptide or NO synthase dilate vessels on stimulation, while those expressing somatostatin drive constriction.²⁶ Similar evidence for neuronal vascular contacts at the capillary level has also emerged in the retina, where gaps in the tessellated processes of Müller glia in the intermediate vascular plexus are occupied by the processes of various neuronal subtypes.²⁷ Comparable fenestrations and neuronal contacts were also seen in neocortical data sets,²⁷ suggesting that direct neuron-capillary contacts are a general feature of central nervous system microvascular organization.

These recent findings add interesting dimensionality to the signaling possibilities in the PVS: beyond signaling factors being released directly from astrocyte endfeet (see below), this also creates opportunities for the release of microglia-derived signaling factors (such as ATP/adenosine,²⁸ COX-1 products,²⁹ or cytokines³⁰), and neuron-derived molecules directly into the PVS from its outer boundary.

Perivascular Macrophages and Fibroblasts

Perivascular macrophages (PVMs) are resident immune cells in the brain that occupy the PVS sandwiched between the glia limitans and the basal lamina of SMCs.³¹ PVMs clear substances from the PVS³² and play a protective, immunoregulatory role during central nervous system infection.³³ Intriguingly, PVMs are predominantly associated with pial arteries and PAs, with SMCs being required for proliferation of these cells,³⁴ whereas PVM association with venules/veins is sparse and is completely absent in the capillary bed.³⁴ Perivascular fibroblasts, in contrast, are found around both PAs and AVs and in the arteriole-capillary transition zone in the initial capillary bed.^{35,36} These cells deposit the ECM coating blood vessels during development and, like PVMs, occupy the PVS between the basal lamina of SMCs and the abutting astrocyte endfeet³⁷ and are somewhat mobile within this space.³⁶

The PVS Around AVs

A structured PVS is often absent around AVs, although collagen fibers can be observed in this space.¹³ Like PAs,

AVs are surrounded by a layer of BM which is then surrounded by an astrocytic endfoot sheath. The AV PVS has been proposed as a key drainage pathway for glymphatic clearance³⁸ (alongside several other pathways including intramural periarterial drainage³⁹ and dural/perineural lymphatic routes⁴⁰). Neurovascular signaling could in principle modulate this compartment, but the influence of signaling here is limited by the properties of venous SMCs, which lack organized contractile smooth muscle,⁴¹ and therefore lack the active wall motion through which arteriolar and capillary signaling shapes perivascular flow. Consequently, whatever the venular PVS contributes to clearance, its relevance to active flow control—the core interest for this review—is minimal, and thus we focus our discussion primarily on the arteriolar and capillary segments of the vascular bed.

Collapse of the PVS at the Capillary Level

Around capillaries, there is no true PVS as the endfeet fuse with the endothelial BM.⁴² The BM-filled gap between endfoot and EC membranes (in which pericytes are also enmeshed) is just 40 to 100 nm wide at this location¹⁹ (Table 1). This loss of true PVS at the capillary level is, we propose, why transmitted electrical signaling comes to dominate here: with diffusible mediators highly locally constrained by the fused BM, inward rectifier potassium (K⁺; Kir2.1) channel-mediated retrograde hyperpolarization conducted from capillaries to upstream arterioles⁴³ provides a convenient mechanism—indifferent to the collapse of the fluid compartment—for sending signals over long distances to upstream vessels covered in contractile cells that can control the diameter of the vessel and thus blood flow to the capillary bed.

The BM and ECM as a Biophysical Filter

Importantly, the BM and surrounding ECM are not inert scaffolding but are the physical medium through which every signaling mediator must diffuse, and their molecular architecture sets the size, charge, and hydration constraints that determine which species can move freely and which are hindered or excluded. The core of the BM

is built from 2 independently self-assembling polymer networks—a laminin sheet and a collagen IV network—mutually cross-linked by nidogen and by the heparan sulfate proteoglycans perlecan and agrin.⁴² Fibronectin and matricellular proteins, such as SPARC (secreted protein acidic and rich in cysteine) and the fibulins, are also present as minor components.⁴² Each of these constituents contributes a distinct physical property, and it is their superposition, rather than the characteristics of any single molecule, that determines filtering effects.

The collagen IV network is the principal determinant of mechanical rigidity and coarse mesh size. Collagen IV assembles into an irregular polygonal lattice, and because it does not bundle into thick fibrils like the fibrillar collagens, it instead forms an open but mechanically stiff meshwork whose interstices set an approximate size and shape cutoff.⁴⁴ The laminin network, anchored to integrins and dystroglycan on the abluminal endothelial and mural cell surfaces, functions as the cell-interfacing layer.⁴⁵ Laminin isoform identity varies systematically along the vascular tree (below) and governs which cells adhere where, and nidogen is a monomeric glycoprotein that is proposed to link the laminin and collagen IV networks together.⁴⁶ Fibronectin, where present, adds tensile reinforcement.⁴⁷

The fixed charge of the BM resides overwhelmingly in the sulfated glycosaminoglycan side chains of the heparan sulfate proteoglycans.⁴⁸ Perlecan and agrin carry long heparan sulfate chains that establish a high local density of immobile negative charge and contribute to the hydrated, gel-like state of the matrix.^{48,49} This hydration is functionally important by providing a larger water-filled volume fraction and therefore a higher effective diffusivity for small neutral solutes than a collapsed dehydrated matrix otherwise would. Interestingly, the fixed negative charge may bias the partitioning of charged species as a solute threads its way through the hydrated glycosaminoglycan mesh, providing the basis for an electrostatic-partitioning argument. Here, cationic solutes (primarily K⁺ ions in the context of neurovascular signaling) would be favored and concentrated by a Donnan-like equilibrium, while anionic solutes would be repelled. These possibilities are explored in more detail below.

This filter is not uniform along the vascular tree. At the pial and PA levels, the fluid-filled PVS is bounded internally by the vascular (pial/endothelial-associated) BM and externally by the glia limitans BM, with the smooth muscle layer residing intramurally within the vessel wall (rather than within the space itself). At this level, the parenchymal BM uniquely comprises 2 apposed layers—a pial-derived laminin $\alpha 1^+$ BM and an astrocyte-derived laminin $\alpha 2^+$ BM. As the PA dives and branches toward the capillary bed, the pial-derived contribution is progressively lost and laminin $\alpha 1$ disappears.⁵⁰ At the capillary, the endothelial and astrocytic membranes fuse into a single shared BM,⁴² eliminating the intervening low-resistance perivascular compartment. EC laminin

isoform expression persists through this transition, with the endothelial $\alpha 4$ chain distributed uniformly along the tree, while $\alpha 5$ is present in the endothelial BM of pial vessels, PAs, and capillaries but becomes patchy at the postcapillary AV.^{42,51}

The signaling implication of these compositional differences is that the same mediator will experience a qualitatively different barrier depending on where it is released. At the arteriolar level, 2 thin BMs flank an open PVS, so even more hindered anionic mediators will have access to a relatively low-resistance compartment for wider distribution. At the capillary, by contrast, the fusion of the 2 membranes into 1 shared BM removes that space and imposes the mesh directly between the endothelium and the cells of the parenchyma, which may amplify the size- and charge-dependent discrimination of the BM proposed above. This is the structural basis for expecting that freely diffusing signals (K⁺, NO) will retain their effectiveness along the tree while anionic lipid mediators will become increasingly restricted in their effectiveness in the capillary bed.

NEUROVASCULAR MEDIATORS AT PLAY IN THE PVS

Neuronal activity couples tightly to elevations of local blood flow to support the metabolic needs of the tissue. The underlying neurovascular coupling (NVC) mechanisms of this functional hyperemia are typically viewed through the lens of cell-cell interactions throughout the neurovascular unit. Less attention has been given to how the environment of the PVS shapes neurovascular signaling, but as this is a confined micro/nanodomain environment, even minor changes in mediator levels can lead to dramatic concentration effects and drive signaling outcomes. Space limitations preclude a comprehensive discussion of all proposed NVC mediators, and so in the following sections we briefly expand on example mediators with distinct properties that influence their behavior and contributions to NVC and provide key known aspects of their mechanisms.

K⁺ IONS

Neuronal action potentials result in a rapid efflux of K⁺ during the repolarization phase, which can directly raise extracellular K⁺ concentrations in the local extracellular space to engage vasodilatory mechanisms on nearby vascular cells.^{52,53} K⁺ can also be released from astrocytic endfeet via BK channels⁵⁴ and potentially other means. These increases in extracellular K⁺ activate Kir2.1 channels found on capillary and arteriolar ECs and vascular SMCs. Kir2.1 activation subsequently hyperpolarizes the SMC membrane (being transmitted over long distances and into SMCs via EC-EC and ultimately EC-SMC gap junctions in the case of hyperpolarizations arising in capillary ECs),

leading to closure of SMC VDCCs (voltage-dependent Ca^{2+} channels), arteriolar vasodilation, and increased local blood flow.^{43,54–56} The endfeet play an important role in K^+ clearance from the extracellular space, where astrocytic Kir4.1 and the Na^+/K^+ ATPase have been implicated in rapid K^+ uptake,^{57,58} and thus will influence the off-kinetics of hyperemic signaling. Accordingly, the endfeet play a doubly significant role in local K^+ signaling to control local blood flow: Not only do they provide a geometric boundary to shape signaling, but they may also define kinetics by operating as both potential source and sink for K^+ .

NO

NO is a central mediator in NVC and is thought to originate primarily from nNOS (neuronal NO synthase)-positive interneurons.⁵⁹ Postsynaptic Ca^{2+} influx activates calmodulin, which then stimulates nNOS to synthesize NO,⁶⁰ which, as a gas, readily crosses the PVS. At least 2 mechanisms contribute to the vasodilatory effect of NO. First, on arriving at SMCs, NO stimulates sGC (soluble guanylate cyclase) to promote the conversion of guanosine triphosphate to cyclic guanosine monophosphate, which in turn activates PKG (protein kinase G). NO-PKG signaling stimulates large-conductance Ca^{2+} -activated K^+ (BK) and ATP-sensitive K^+ (K_{ATP}) channels to induce membrane hyperpolarization.⁵⁶ This closes VDCCs and lowers intracellular Ca^{2+} , promoting smooth muscle relaxation and vasodilation.⁵⁶ Second, NO occupies the heme moiety of the cytochrome P450 epoxygenase CYP4A/4F ω -hydroxylase to suppress 20-HETE synthesis, thereby relieving 20-HETE-dependent constriction, which is mediated by protein kinase C-dependent inhibition of BK channels, membrane depolarization, and VDCC Ca^{2+} entry.^{61–64} These dilatory effects of nNOS-derived NO are known to have a central role in functional hyperemia, with nNOS blockade resulting in an average 64% reduction in the neurovascular response.⁶⁵

In parallel to nNOS, NO from capillary ECs can relax nearby ensheathing pericytes in the arteriole-capillary transition zone,⁶⁶ but it is currently unclear whether NO acts on thin-strand pericytes in the deep capillary bed.

ADENOSINE

NVC also involves purinergic signaling via adenosine. Activity-dependent ATP release from neurons and astrocytes is rapidly hydrolyzed in the extracellular space by ectonucleotidases—principally the ecto-nucleoside triphosphate diphosphohydrolases and the ecto-nucleotide pyrophosphatase/phosphodiesterases—generating ADP and AMP, which are then converted to adenosine by CD73/ecto-5'-nucleotidase.^{67,68} Adenosine can also be released from neurons during synaptic transmission⁶⁹ and then activates G_s -linked adenosine A_2Rs (A_2 receptors) that stimulate AC (adenylyl cyclase) to generate

cAMP in target cells, such as pericytes, ECs, and SMCs.⁶⁸ This cAMP rise activates PKA (protein kinase A)-mediated phosphorylation of K^+ channels such as K_{ATP} , leading to enhancement of their function.⁷⁰

PROSTAGLANDIN E_2

Arachidonic acid liberated from membrane phospholipids by Ca^{2+} -dependent cPLA₂ (cytosolic phospholipase A₂) is metabolized by cyclooxygenases to prostaglandin H_2 , which is then converted to prostaglandin E_2 (PGE_2) by microsomal or cytosolic PGE synthases.⁷¹ PGE_2 then activates vascular EP (E-prostanoid) receptors but has been found to elicit varied vasodynamic responses depending on the vessel compartment and EP receptor subtype in question. At the capillary level, direct application of PGE_2 onto capillary endothelium results in upstream arteriolar dilation via EP₁-mediated intracellular Ca^{2+} elevation, engaging a vasodilatory hyperpolarizing signal that propagates towards the capillary-feeding arteriole,⁷² while direct arteriolar stimulation of EP₁ induces G_q -mediated Ca^{2+} -dependent vasoconstriction.⁷³ A role for EP₄-mediated relaxation of ensheathing pericytes around capillaries to induce dilations has also been described.⁷⁴

EPOXYEICOSATRIENOIC ACIDS

Astrocytic cytochrome P450 epoxygenases are responsible for the formation of membrane-permeant epoxyeicosatrienoic acids (EETs) from arachidonic acid.⁷⁵ Nanomolar concentrations of EETs act via at least 2 mechanisms to increase BK channel activity in vascular SMCs.⁷⁶ First, there is growing evidence for a high-affinity EET-binding GPCR (G protein-coupled receptor), though the nature of this putative receptor has yet to be elucidated.⁷⁷ It is proposed that G_s -mediated signaling via the EET receptor (sustained by ADP-ribosylation of the $\text{G}\alpha_s$ subunit⁷⁸) activates AC, leading to cAMP elevation, PKA engagement, and activation of K^+ channels such as K_{ATP} and BK.⁷⁷ A second mechanism proposes that EETs activate TRPV4 (transient receptor potential vanilloid 4) channels, which in turn increase ryanodine receptor-mediated Ca^{2+} spark frequency, promoting BK channel activity.⁷⁹ Although the precise signaling processes remain under active investigation, current evidence firmly positions EETs as important vasodilatory mediators of NVC.

THE BIOPHYSICS OF THE PVS DETERMINE MEDIATOR CONCENTRATION AND DYNAMICS

The brain extracellular space is a vast interstitial reticulum accounting for $\approx 20\%$ of brain volume.^{80,81} The dimensions of these spaces place constraints on the effective diffusion coefficients of molecules that do not

readily pass through membranes (eg, K^+), reducing them on average by about 2.5-fold.^{80,81} As a result, the concentration of signaling molecules that can be achieved away from a release site drops off nonlinearly and rapidly with distance,⁸⁰ raising a barrier for effective neurovascular communication. One solution to this problem lies in the PVS around vessels providing a zone into which molecules can be released and rapidly concentrated to optimize signal transduction.

As discussed above, the BM found throughout the PVS essentially functions as a molecular sieve with both size⁸² and charge⁸³ selectivity, which will shape signaling events. For small vasoactive molecules, the impact of the BM will be minimal. Gaseous mediators such as NO, ions such as K^+ (on the order of 0.1 nm including hydration), and molecules such as PGE_2 , EETs, and adenosine (with estimated Stokes-Einstein radii^{84,85} of ≈ 0.4 – 0.6 nm) will rapidly permeate the local PVS upon their release. Focusing on the capillary level, crossing the 40 to 100 nm gap between membranes will take microseconds for molecules of these diminutive sizes. For larger molecules such as growth factors, the situation becomes more complex, and the PVS provides a reservoir effect. Considering VEGF (vascular endothelial growth factor) as an example, this molecule binds to heparan sulfate proteoglycans,⁸⁶ an interaction that has the dual effect of slowing its diffusion across the PVS and concentrating molecules within this compartment. As a result, the dynamics of VEGF signaling will be smoothed over time, with slower on and off kinetics than smaller molecules. Accordingly, in scenarios requiring rapid signaling to acutely modulate blood flow, small molecules such as K^+ , NO, and adenosine are ideal. For tasks which require signaling over long time frames—for example, modification of membrane composition to modulate signaling strength—larger molecules may be preferable as their interactions with the BM will extend their duration of effect (Figure 2).

Another interesting consideration is the charge effects of the BM heparan sulfate proteoglycans. The strong negative charge on these proteins could have Donnan-like effects which attract and help to concentrate K^+ ions, effectively creating an electrostatic partitioning of ions into this space. Indeed, such a phenomenon has been described for chloride (Cl^-) ions around neurons,⁸⁷ and so an analogous process could be important in amplifying K^+ concentrations around capillaries to a level sufficient to evoke Kir2.1 signaling and electrical communication in response to neuronal activation. Probes for visualization of ion distribution and direct measurements of BM charge are needed to analyze these possibilities in detail (Figure 2).

A further important consideration regarding concentration of molecules in the PVS compartment is receptor affinity, and a relevant example is provided by adenosine: $A2_B$ receptors have been implicated in

adenosine-mediated functional hyperemia, with $A2_A$ blockade sparing and $A2_B$ blockade attenuating the whisker-evoked cerebral blood flow response.⁸⁸ The lower adenosine affinity of $A2_B$ ⁸⁹ is consistent with engagement during the later, sustained phase of the NVC mechanism. This contrasts with the higher-affinity $A2_A$ receptor, which has been identified on smooth muscle⁹⁰ and pericytes,⁷⁰ which will respond more rapidly to local adenosine release. These differential receptor characteristics underscore an important point: not only will the time taken to generate/release and concentrate adenosine in the PVS be a key determinant of its contribution to NVC, but the specific receptor subtype expressed by the target cell plus the time taken for signal transduction will also influence when a molecule can contribute to vasomotor responses (Table 2).

In defiance of the tortuosity of the extracellular space, NO and K^+ are able to traverse limited distances when not released directly into the PVS. NO readily crosses membranes and therefore will diffuse equally in all directions from its source of release, creating a sphere of influence on signaling processes in local cells. Although the physical geometry of the extracellular space thus does not impose hard limits on NO signaling, it is limited by its short half-life and the range of interacting partners that will bind NO molecules as they traverse the local milieu. This is thought to constrain the effective communication distance of NO from any point source to a length constant on the order of low single-digit micrometers,¹⁰⁴ beyond which its concentration will fall to a level too low to achieve effects above noise. Further, while NO diffuses rapidly, its activity will be delayed by its transduction latency, suggesting it will be slower than K^+ , which has only to reach Kir channels in the extracellular membrane of the target cell to initiate channel activity and electrical signaling (Table 2). This ordering is supported by recent data suggesting that NO does not participate in the initiation of the hemodynamic response to brief sensory stimulation.¹⁰⁵

K^+ ions do not pass membranes alone, and thus these will have to diffuse through tortuous extracellular space if not released directly onto capillaries. However, recent modeling that accounts for parenchymal tortuosity assessed the effects of various widths of capillary PVS and gaps between endfeet as potential impediments to K^+ and found that these do not pose a significant barrier to rapid K^+ accretion. Even at distances of 15 μm from the vessel wall (the average distance of a neuronal cell body from its nearest capillary¹⁰⁶), release from neurons can raise capillary PVS K^+ by several mM within several hundred milliseconds, dropping to tens of milliseconds if released very nearby.⁵³ Tetramethylammonium ion diffusion data in the extracellular space also suggest little impedance to monovalent cation diffusion, in agreement with modeling data.¹⁰⁷ Thus small, aqueous molecules are ideally suited for rapid neurovascular communication in the PVS.

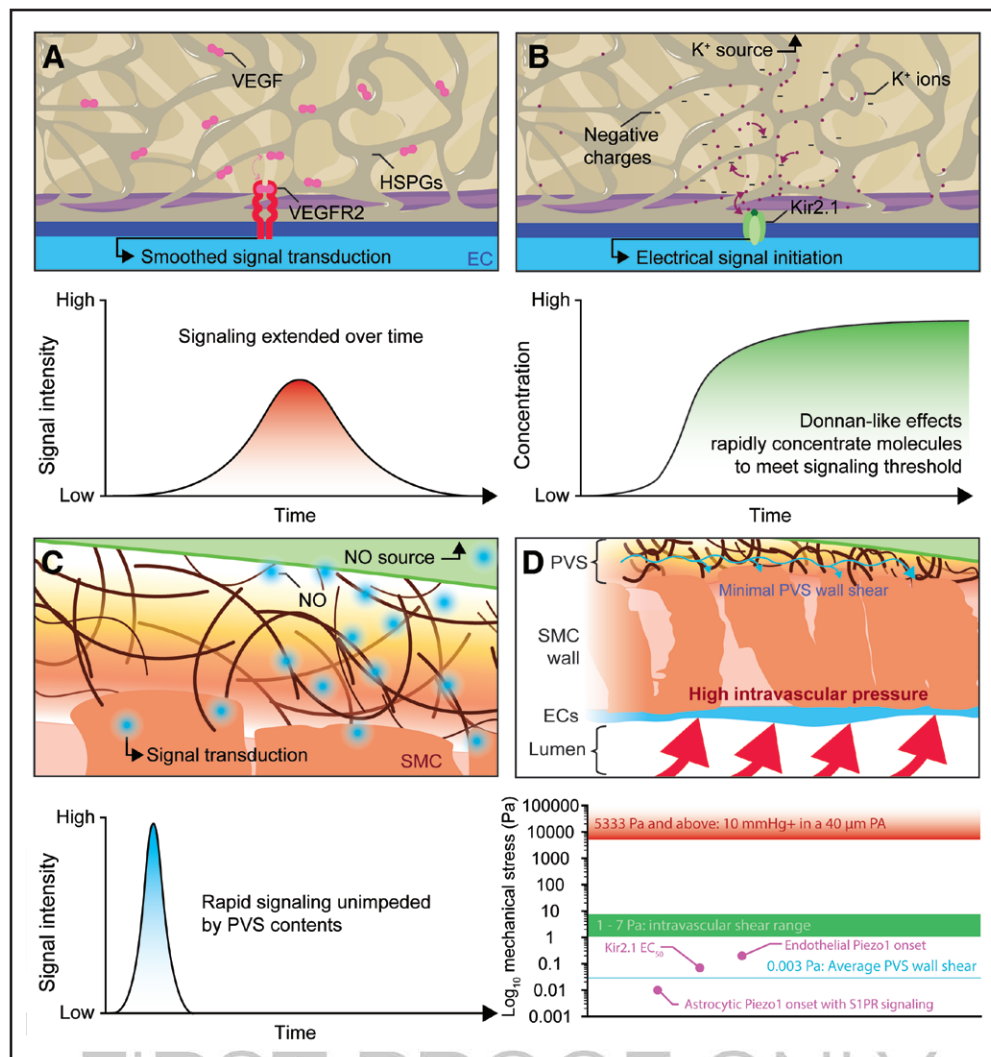


Figure 2. The impact of perivascular space (PVS) biophysics on neurovascular signaling dynamics.

A, Binding of large molecules such as VEGF (vascular endothelial growth factor) to heparan sulfate proteoglycans (HSPGs) may create a reservoir effect that maintains signaling concentrations for long periods and extends effects over long durations. **B**, Donnan-like charge effects may also help concentrate small, charged molecules such as K^+ to achieve signaling concentrations. **C**, The PVS is no impediment to gaseous mediators such as nitric oxide (NO), enabling rapid neurovascular communication, making signal transduction the most likely kinetic bottleneck. **D**, The smooth muscle is optimized to sense large intravascular pressures and is unlikely to respond to the minimal wall shear imparted by fluid moving through the PVS. The chart compares the forces imparted by intravascular pressures in penetrating arterioles (PAs) and pial arteries to the known range of intravascular and PVS glymphatic shear forces on a common axis. Also included are the pressure sensitivities of example mechanosensing elements. Kir2.1¹⁷³ and unmodified Piezo1¹⁷⁴ as found in endothelial cells (ECs) are highly sensitive, but are unlikely to sense the minimal shear from glymphatic flux. Recent evidence has suggested that sphingosine-1-phosphate receptor signaling may increase the sensitivity of astrocytic Piezo1 to an even greater degree¹³⁰ to bring it into the range of shear forces imparted by fluid moving through the PVS. Although the plotted quantities are of physically distinct types (transmural-pressure-derived circumferential wall stress, fluid shear stress, and membrane-tension-equivalent activation thresholds), all carry units of force per unit area and are displayed on a common logarithmic axis to enable comparison of their relative magnitudes. SMC indicates smooth muscle cells.

In contrast to the rapid actions of K^+ and NO, several features of PGE_2 and EETs suggest that they will operate on longer timescales. Indeed, the lipophilicity of EETs and PGE_2 complicates the argument for their release into the extracellular space and simple diffusion to adjacent membranes as their primary mechanism of action, as either molecule will more readily partition into membranes and move laterally within that 2-dimensional plane to achieve signaling effects.^{108,109} This view is reinforced by data indicating that the binding

site for PGE_2 on EP receptors is located within the lipid bilayer itself,¹¹⁰ and data showing that EETs partition into lipid bilayers to exert signaling effects on ion channels.¹¹¹ Given this preference for lipid environments, and the lack of direct membrane contacts between the endfoot layer and ECs or pericytes, this argues against a model in which these mediators are released into the PVS to diffuse to the vasculature to exert their effects. Indeed, the aqueous environment of the PVS will exert pressure on any molecules released from cells to turn back into

Table 2. Kinetics of Neurovascular Coupling Mediators in the PVS

Mediator [phase]	Half-life	Receptor-effector kinetics	Interpretation and notes	References
K ⁺ [fast]	No degradation; cleared by astrocytic Kir4.1 uptake + Na ⁺ /K ⁺ -ATPase; clearance $t_{0.5}$ estimated at 3 s from data in ref ⁵⁸	Direct Kir2.1 gating (μ s - ms) + electrotonic propagation to arteriole (>2 mm/s)	Minimal latency to onset of signaling, which should track the rise in local [K ⁺] _i closely; that is, neural firing itself	43,58,80,91
NO [intermediate]	0.64 s cortex; 90 ms to >2 s perivascular (O ₂ ⁻ and distance-dependent)	sGC → cGMP → PKG signaling takes ≈0.5–1 s	Limiting steps are likely cGMP buildup and further signaling cascade steps; the short lifetime of NO limits its spatial spread	92–96
Adenosine [intermediate]	~2–3 s in vivo.	A _{2A} /A _{2B} G _s activation onset occurs in ≈0.3–0.5 s; cAMP accumulates over substantially longer time scales	cAMP accumulation and further signaling cascade steps are likely to be rate-limiting	97,98
EETs [slow]	n.d.	Time to TRPV4/BK activation (currently unknown); astrocyte Ca ²⁺ -dependent, which lags dilation onset during NVC	Likely limited by astrocyte Ca ²⁺ response and time for synthesis from AA by PLA ₂	77,79,99–101
PGE ₂ [slow]	n.d. for brain	EP ₁ G _q GPCR activation occurs in 0.3–0.5 s; PLC-mediated Ca ²⁺ release occurs over longer time scales; EP ₄ G _s activation occurs in ≈0.3 to 0.5 s; cAMP accumulates over longer time scales	Limiting steps are likely to be second messenger build-up and subsequent signaling steps, plus time for synthesis from AA by PLA ₂	98,101–103

Rate-limiting factors are likely the accumulation of second messengers (eg, cGMP/cAMP) and synthesis of signaling molecules (eg, PGE₂/EETs). Diffusion time across the PVS was deliberately omitted as it is not a distinguishing variable (Figure 4; Supplemental Materials). Indeed, for every mediator, the PVS is crossed in ≤10 ms, much faster than the ≈0.5 to 1 s onset of functional hyperemia. Therefore, we conclude that diffusion is permissive and will not impact response kinetics. AA indicates arachidonic acid; EET, epoxyeicosatrienoic acid; n.d., no data; NVC, neurovascular coupling; PGE₂, prostaglandin E₂; PLA₂, phospholipase A₂; PVS, **perivascular space**; and TRPV4, transient receptor potential vanilloid 4.



the membrane of the producing cell, meaning release events will strongly favor autocrine actions as opposed to trans-PVS signaling. However, it must be noted that mediators such as these could undergo carrier-mediated or protein-facilitated transport. PGE₂ is anionic and membrane-impermeant, requiring carrier-mediated efflux for release,^{112,113} whereas the lipophilic EETs cross membranes freely but depend on fatty-acid-binding proteins for intracellular trafficking to their targets.^{114,115} Given that aqueous diffusion of an unchaperoned highly lipophilic molecule across the PVS is biophysically unfavorable on the timescale of NVC, the most parsimonious weighing of these factors implies 3 main possibilities: (1) Upstream signaling events drive lipid mediator production within ECs, pericytes, or SMCs themselves, which then act in an autocrine fashion to trigger hyperemic signaling; (2) Lipid mediators act within the membrane of their cell of origin to trigger the release of other hydrophilic molecules, as has been described for EETs signaling within the astrocytic endfoot membrane to trigger BK activation and K⁺ release into the PVS¹¹⁶; or (3) transport proteins within the ECM are crucial for transfer of lipid mediators to their signaling target.

For PGE₂, the constraints these considerations place on its range are likely important to unify observations of its opposing actions at different levels of the vascular tree (ie, EP₁-mediated constriction of arterioles,⁷³ EP₄-mediated dilation of ensheathing pericytes,⁷⁴ and EP₁-mediated transmitted hyperpolarization from capillaries,⁷² detailed above). If PGE₂ were able to diffuse freely, these effects may cancel 1 another out, but the limitations on trans-PVS PGE₂ diffusion may actually favor

these differential outcomes by helping to concentrate its effects at the desired receptor/cell type in the signaling cell's immediate neighborhood.

A further issue impacting the contribution of these molecules to blood flow response to neuronal activation is the time it takes for their synthesis, in addition to any delays introduced by signal transduction. Using EETs as an example, synthesis canonically involves initiation of Ca²⁺ release, activation of PLA₂, generation of arachidonic acid from membrane phospholipid pools, and conversion of arachidonic acid to EETs via cytochrome P450 enzymes.^{56,117} This multistep process, plus the need to appropriately concentrate the mediator at its site of action and for signal transduction effects described above to ensue, naturally introduces a time delay until biological effects are observable. This is supported by laser Doppler flowmetry and imaging experiments in which the initial fast response of the hyperemic response was not affected by EET inhibition, but the delayed, sustained response to continued stimulation was reduced.^{88,99} Recent evidence indicates that EETs are released during slow Ca²⁺ elevations in astrocytes, the generation of which adds a further lag to the onset of their effects.⁹⁹

Accordingly, the biophysical properties of the PVS argue for a temporal hierarchy that separates mediators according to the constraints that these properties place on their activity. Here, fast initiation is likely mediated by K⁺ (immediate release, no need for synthesis, rapid electrical action) while the sustained response is likely supported by mediators requiring enzymatic synthesis, membrane partitioning, and extracellular accumulation

before they can exert their effects (NO will be intermediate, limited by transduction latency; adenosine also intermediate, limited by production/release, concentration, and second messenger build up; and EETs/PGE₂ will be the slowest, limited by synthesis, membrane partitioning, and accumulation). Figure 3 provides a time course illustrating this proposed temporal hierarchy, and detailed mediator kinetics are explored in greater depth in Table 2.

GLYMPHATIC FLUX THROUGH THE PVS—A SOURCE FOR MECHANICAL INPUT?

Mechanical forces imparted by passing red blood cells generate luminal shear forces that can modify arteriolar diameter through endothelium-dependent mechanisms. Intravascular pressure also controls tone and autoregulation, and thus mechanical inputs play important roles in determining SMCs' contractile state, vessel diameter, and blood flow.¹¹⁸ On the other side of the vessel wall, the glymphatic flux of fluid through the PVS⁸ imparts abluminal mechanical force that invites the hypothesis that this might exert sufficient shear on the vessel wall to be transduced by mural mechanosensors, thereby modifying vascular tone and, in turn, flux itself, constituting a potential feedback loop. Are the abluminal mechanical forces at play here sufficient to trigger such signaling?

Glymphatic flux at the level of pial arteries on the brain surface is thought to involve perivascular pumping, where arterial wall motion aligned with the heartbeat provides the impetus to propel fluid along ovoid channels

around vessels.^{15,119,120} A question under active investigation is whether this pumping mechanism is sufficient to fully explain directional glymphatic flux into the brain, or whether it instead produces oscillatory motion of fluid in the PVS with an additional pressure gradient required to produce brain-directed motion.¹²¹

As arteries transition to PAs, it has been proposed that the endfoot layer provides a valve-like pumping mechanism to favor flux of glymphatic fluid into the brain.^{122,123} Here, the dilation of the PA will narrow the PVS and thereby compress the extracellular fluid. This will increase pressure in the PVS, which is proposed to expand the gaps between endfeet, increasing their permeability and pushing fluid into the brain parenchyma. When the arteriole then returns to its resting diameter or constricts, its decrease in diameter will expand the PVS and thereby drop local pressure. The endfoot gaps will simultaneously tighten and create a path of least resistance for more fluid to be drawn in along the PA from the pial reservoir at the brain surface.^{122,124}

At the capillary level, the absence of a true PVS and presence of the BM imposes high hydraulic resistance. This is thought to shift the dominant mode of fluid movement from advective to diffusive^{119,125} and thus the mechanical forces on the abluminal surface of the vessels are likely to be very small and thus will not make a major contribution to signaling events in this compartment.

In the venous circulation, the respiratory cycle is thought to be the dominant driver of fluid movement. During inspiration, intrathoracic pressure becomes negative, reduces venous blood volume, and creates a draw for

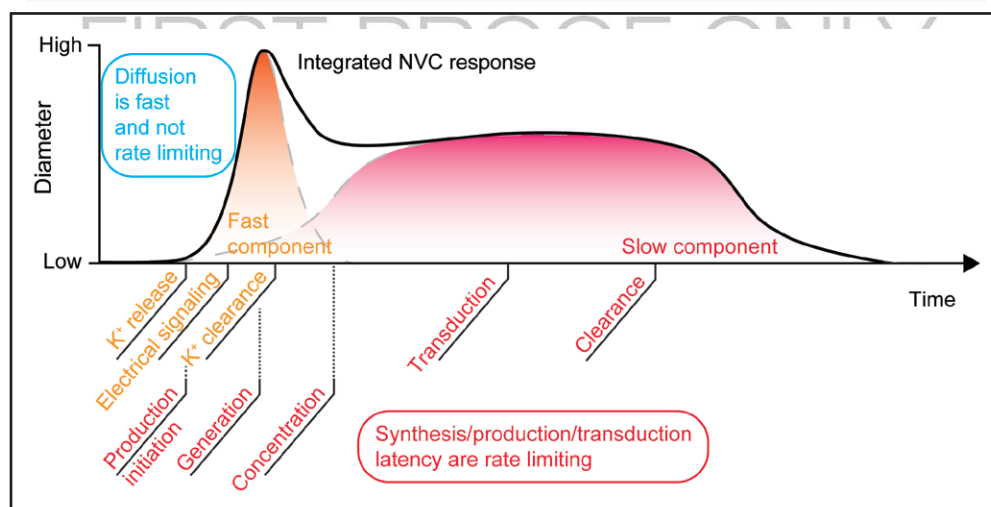


Figure 3. Proposed temporal hierarchy for biphasic neurovascular coupling (NVC) expressed as penetrating arteriole diameter over time.

The integrated NVC response consists of an initial fast component, which we propose is likely mediated by K⁺ ions followed by nitric oxide release and adenosine concentration, followed by a slower component in which epoxyeicosatrienoic acids and likely prostaglandin E₂ also play a role. Diffusion across the perivascular space is very rapid, occurring within milliseconds, and thus does not place limitations on mediator signaling. Instead, the time taken for synthesis and production of mediators, combined with the time it takes for signal transduction, affects the time it takes for a mediator to contribute to the vasomotor response. For K⁺ in particular, delays are minimized by the fact that it is immediately released during action potentials and diffuses to the vascular wall in milliseconds, and its action on Kir2.1 channels to initiate membrane currents is very rapid. The kinetics of specific mediators considered in this review are explored in greater detail in Table 2.

further fluid accumulation, while expiration has the opposite effect of narrowing the PVS and propelling fluid.¹¹⁹

The foregoing suggests that the most likely point at which the motion of fluid in the PVS could make a contribution to signaling is via input to mechanosensors in the SMCs¹²⁶ that directly interface with the PVS at the arterial and arteriolar level. However, the forces imposed by glymphatic movement are almost vanishingly small. Recent work has inferred in vivo PVS flow-associated shear stress at ≈ 0.003 Pa (ie, 0.03 dyne/cm²).¹²⁷ This is orders of magnitude lower than intraluminal shear stress experienced by the endothelium (in the range of 1 – 7 Pa¹²⁸) and, even more starkly, converting intraluminal pressure of 40 mmHg in a 35 μ m PA with wall thickness of 4 μ m¹²⁹ yields an estimated circumferential wall stress of around $23\,000$ Pa— >7 -million-fold larger than PVS longitudinal shear. The mechanosensing machinery in the pial artery and PA SMCs is optimized for circumferential sensing of this intraluminal pressure, not shear along the longitudinal axis of the vessel as in the case of PVS fluid flow. Given the stark differences between the size of these forces, it is unlikely that the latter contributes meaningfully to signaling in this space (Figure 2).

Although the SMC mechanosensors will not be impacted by glymphatic mechanical forces, astrocytes may contain modifications that allow them to sense glymphatic flux. Indeed, a recent study outlined a mechanism by which the sensitivity of Piezo1 channels in astrocytic endfeet may be augmented via sphingosine-1-phosphate receptor signaling to bring it into the range of glymphatic flux forces.¹³⁰ It is also noteworthy that dilation of vessels will compress the PVS, causing a spike in pressure (as described for the putative astrocyte valve mechanism¹²²)—a distinct case in which local PVS forces could enter the response domain of SMC or astrocytic mechanosensors. More speculatively, it is not known whether mechanical forces from the lumen can be transduced through capillaries and into the PVS—capillaries are incredibly narrow, on the order of 6 μ m in diameter in humans^{131,132} and 2 to 5 μ m in mice,^{133,134} which means that red blood cells (≈ 8 μ m in humans¹³⁵ and ≈ 6 μ m in mice¹³⁶) squeeze to enter them, exerting an irregular, pulsatile, and dynamic mechanical force onto the walls of the capillaries as they pass. This is sensed by endothelial Piezo1 channels and has roles in mechano-feedback to control blood flow.¹³⁷ It is interesting to ask whether, given the exceptional thinness of capillary ECs in non-nuclear regions,¹³⁸ these physical forces imparted by passing red blood cells are completely buffered by the ECs or whether they may also be transferred to the capillary-adjacent, BM-filled PVS, which could then mechanically impact pericytes and astrocytes. PVMs and perivascular fibroblasts also both express Piezo1,^{139,140} suggesting that their activity could be modulated by changes in glymphatic flux, which could potentially link such forces to immune state and ECM composition, respectively.

Thus, shear from glymphatic flux is very unlikely to directly act on SMCs, but there may be restricted cases (dilations at the level of the arterioles, red blood cell flux at the level of capillaries) in which mechanical forces do influence PVS geometry and signaling.

HYPERTENSION AS A SYSTEMIC DISRUPTOR OF PVS SIGNALING

Chronic elevated blood pressure (hypertension, defined as systolic pressure above 130 mmHg and diastolic pressure over 80 mmHg¹⁴¹) has a range of impacts on the vasculature which impinge on the PVS. Large blood vessels are compliant structures, with arteries elastically deforming in response to pressure waves generated by pumping of the heart. This elastic compliance is supported by elastin and collagen deposited around arteries and arterioles: proteins which store and release mechanical energy and provide tensile strength, respectively. ECM remodeling, including elastin fragmentation and collagen deposition, is a hallmark of hypertensive pathology in the periphery.^{142,143} Because perivascular fibroblasts deposit ECM, they are a likely candidate for reshaping the PVS in pathological conditions by, for example, depositing excess collagen leading to vessel stiffening and a reduction in vascular compliance. Indeed, this is supported by observations of collagen deposition in the brains of spontaneously hypertensive rats.¹⁴⁴ Intriguingly, this same group noted that CD206+ PVMs were enriched with mRNA for the *Col1a1* gene during hypertension, suggesting PVMs also contribute to collagen deposition and PVS ECM remodeling. Vascular stiffening from collagen deposition could thus blunt the arterial distention that drives fluid flow in the PVS, disabling glymphatic clearance and connecting cardiovascular pathology to brain injury.¹⁴⁵ Loss of blood vessel compliance will also make vasculature more prone to pressure-wave injury.¹⁴⁶

Blood-brain barrier (BBB) disruption is another hallmark of hypertension. NOX2 (nicotinamide adenine dinucleotide phosphate oxidase 2) generates superoxide that can result in vascular injury in this context. Indeed, in mouse models of hypertension, PVMs were found to respond directly to extravasated Ang II (angiotensin II) via AT₁R (Ang II type 1 receptor) signaling, triggering reactive oxygen species production via PVM-NOX2 activity, leading to subsequent vascular oxidative injury.^{147,148} This injury reduces NVC, blunting blood flow increases to active brain regions and worsens BBB opening associated with hypertension. Depletion of PVMs themselves, PVM AT₁Rs, or PVM NOX2 rescues NVC¹⁴⁷ and, intriguingly, depletion of AT₁Rs from vascular ECs also prevents the initial BBB injury which results in extravasation of Ang II.¹⁴⁸ This suggests that PVMs and ECs synergistically drive BBB disruption and pathology in hypertension. A further angiotensin-independent route

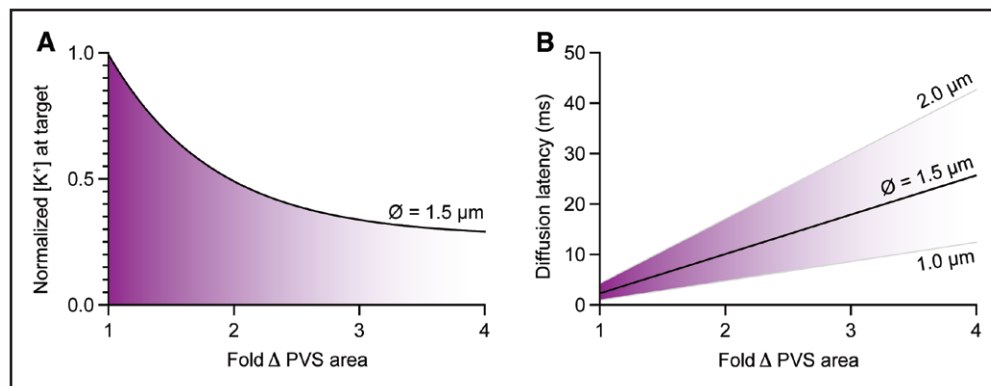


Figure 4. Effect of perivascular space (PVS) expansion on K⁺ signaling to arteriolar smooth muscle.

Astrocyte endfeet sheath almost the entire vessel surface, and thus here K⁺ release was modeled as a uniform source distributed around the vessel. **A**, Delivered K⁺ concentration at the vessel wall, normalized to baseline, as a function of PVS cross-sectional area expansion. As expected, doubling the PVS area halves K⁺ concentration at the vessel wall. **B**, Latency for K⁺ to cross to the vessel vs PVS area expansion for baseline gaps of 1.0 to 2.0 μm (midpoint 1.5 μm). Latency remains ≤30 ms even at maximal expansion, which is 2 to 3 orders of magnitude below the ≈0.5 to 1 s onset timescale of functional hyperemia, so diffusion is minimally impacted. Thus, dilution, not delay, is the functionally dominant consequence of PVS expansion. See [Supplemental Materials](#).

to neurovascular and cognitive dysfunction is provided by dural meningeal T-cell IL (interleukin)-17 release signaling to border associated macrophages.¹⁴⁹

A well-documented finding from human neuroimaging is that hypertension also produces an increase in the size of the PVS around arteries and arterioles,⁷ and this change has been used as a clinical marker of cerebral small vessel disease (of which hypertension is a major driver).¹⁵⁰ Evidence of PVS expansion is largely derived from human cohorts in which white matter hyperintensities are also present,^{151–155} leaving the question of whether PVS enlargement precedes or co-occurs with white matter hyperintensities or other signs of parenchymal injury. Recent preclinical studies in spontaneously hypertensive rat and salt-sensitive animal models do show that hypertension is associated with white matter damage^{156,157} and glymphatic impairment,^{158,159} but whether PVS enlargement is an early or damage-independent event remains to be established and is an important direction for future research.

PVS enlargement is a consequence of reduced vascular reactivity¹⁶⁰ and can be explained by the stalling and backflow of fluid causing pooling around these vessel segments, which in turn drives distension.¹⁵ In turn, this PVS enlargement could allow for the accumulation of toxins which cannot be readily cleared by glymphatic pumping in a larger volume and may also lead to further reductions in vascular reactivity by diluting signaling molecules in the PVS. Modeling studies suggest that even a modest expansion in PVS geometry alters flow patterns, diffusion path length, solute transport, and dilution volume, which has stark implications for neurovascular communication.^{161,162} For molecules with a steep concentration-response relationship such as K⁺,⁴³ this could meaningfully impact the ability to engage Kir2.1 channels to a degree that will overcome background conductances to produce membrane hyperpolarization.

Indeed, directly modeling the relationship between arteriolar PVS volume and K⁺ behavior (Figure 4) indicates that a 2-fold expansion of the PVS will drop effective steady-state K⁺ concentration arriving at the vessel wall by ≈50%. For a brief elevation of K⁺, this could dramatically reduce the level of resultant hyperpolarization and blunt vessel dilations.

Barrier dysfunction in hypertension also introduces plasma components, predominantly into the capillary PVS where barrier breakdown is most prominent, such as albumin and fibrinogen.¹⁶³ Albumin binds to a range of active signaling molecules in the blood, and its introduction to the PVS could thus bind and sequester signaling molecules such as PGE₂.¹⁶⁴ The introduction of blood components will also drive further inflammation,¹⁶⁵ and inflammatory mediators are known to affect the expression and function of signaling elements that are critical at this interface. Examples include interleukin-1β and tumor necrosis factor-α driving VDCC overactivity in smooth muscle¹⁶⁶ and transforming growth factor-β1 suppressing endothelial Kir2.1 channel function.¹⁶⁷ Inflammation increases the production of reactive oxygen species in the vascular wall, for example, through overactivity of NOX2.¹⁶⁸ This elevates levels of superoxide, which can react with NO to produce peroxynitrite (ONOO[−])—a superoxide derivative that can act as a mediator of vascular dysfunction in hypertension.¹⁶⁹ This could thus have a double-hit effect on the neurovascular unit: not only does ONOO[−] formation cause cellular damage in its own right, but the attenuation of NO signaling will impair blood flow control and consequently energy delivery.

Unchecked hypertension could, through these processes, lead to a self-amplifying cycle that compounds damage to the neurovascular unit. The initial stiffening of the vascular wall will reduce perivascular pumping¹⁵ and lower the pulsatility-dampening effect¹⁷⁰ that normally

protects the delicate microvasculature. The loss of clearance mechanisms in the PVS will cause the accumulation of toxic molecules such as amyloid β ,³⁸ and breach of the BBB¹⁷¹ will add further molecules to the stagnant PVS, which could drive further signaling disruptions and further increase wall stiffening—overall producing a feed-forward cycle that could contribute significantly to brain disorders. However, the results of a recently conducted clinical trial indicate that intensive blood pressure lowering is capable of at least partially reversing arteriolar PVS expansion measured on MRI,¹⁷² suggesting that management of blood pressure may prevent PVS disruption and that aggressive treatment of existing hypertension may help to correct aberrant PVS signaling and thus protect brain health and function.

ARTICLE INFORMATION

Affiliation

Department of Pharmacology and Physiology, University of Maryland School of Medicine, Baltimore.

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Disclosures

None.

Supplemental Material

Supplemental Material (Modeling Details)

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